

Problem 1. A 2×2 real matrix $A \in \mathbb{R}^{2 \times 2}$ can be thought of as a linear function $\mathbb{R}^2 \rightarrow \mathbb{R}^2$. Tell me the matrices corresponding to the following functions.

- (a) Rotate counterclockwise by 90° .
- (b) Project orthogonally onto the line $y = x$.

This be done directly, using the fact that $A\mathbf{e}_1$ and $A\mathbf{e}_2$ are the columns of A . Or we can remember the general formulas:

$$R_t = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}, \quad P_{\mathbf{v}} = \frac{1}{\|\mathbf{v}\|^2} \mathbf{v} \mathbf{v}^T.$$

Using the angle $t = 90^\circ$ and the vector $\mathbf{v} = (1, 1)$ gives

$$R_{90^\circ} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad P_{(1,1)} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Problem 2. Consider a complex $m \times n$ matrix $U \in \mathbb{C}^{m \times n}$. Prove that

$$U^*U = I_n \iff \langle U\mathbf{x}, U\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle \text{ for all } \mathbf{x}, \mathbf{y} \in \mathbb{C}^n,$$

with respect to the standard Hermitian inner products on $\mathbb{C}^m$ and $\mathbb{C}^n$.

For the $\Rightarrow$ direction, suppose that $U^*U = I$. Then for all $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$ we have

$$\langle U\mathbf{x}, U\mathbf{y} \rangle = \langle \mathbf{x}, U^*U\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle.$$

For the $\Leftarrow$ direction, suppose that $\langle U\mathbf{x}, U\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$. We will give two proofs that $U^*U = I$. In terms of matrix multiplication: We have assumed that $\mathbf{x}^*U^*U\mathbf{y} = (U\mathbf{x})^*(U\mathbf{y}) = \langle U\mathbf{x}, U\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^*\mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$. Letting $\mathbf{x} = \mathbf{e}_i$ and $\mathbf{y} = \mathbf{e}_j$ be standard basis vectors shows that

$$(ij \text{ entry } U^*U) = \mathbf{e}_i^*U^*U\mathbf{e}_j = \mathbf{e}_i^*\mathbf{e}_j = \delta_{ij},$$

and hence $U^*U = I$. In terms of abstract adjoint operators: We have

$$\begin{aligned} \langle U\mathbf{x}, U\mathbf{y} \rangle &= \langle \mathbf{x}, \mathbf{y} \rangle \\ \langle \mathbf{x}, U^*U\mathbf{y} \rangle &= \langle \mathbf{x}, \mathbf{y} \rangle \\ \langle \mathbf{x}, U^*U\mathbf{y} \rangle - \langle \mathbf{x}, \mathbf{y} \rangle &= 0 \\ \langle \mathbf{x}, U^*U\mathbf{y} - \mathbf{y} \rangle &= 0 \\ \langle \mathbf{x}, (U^*U - I)\mathbf{y} \rangle &= 0. \end{aligned}$$

Since this holds for all $\mathbf{x}, \mathbf{y}$ we can take $\mathbf{x} = (U^*U - I)\mathbf{y}$ to get

$$\begin{aligned} \langle (U^*U - I)\mathbf{y}, (U^*U - I)\mathbf{y} \rangle &= 0 \\ \|(U^*U - I)\mathbf{y}\|^2 &= 0 \\ (U^*U - I)\mathbf{y} &= \mathbf{0} \\ U^*U\mathbf{y} &= I\mathbf{y}. \end{aligned}$$

Then since this holds for all $\mathbf{y}$, we conclude that $U^*U = I$. □